

Polynomial Regression Assignment

Geothermal Power Plant Expansion Project

Roll No.: BT2024136

All code is available in the GitHub repository:

https://github.com/yashvi2706/BT2024136_ML_Assignment1.

1 Introduction

This report describes polynomial regression models for two personalised datasets. **var1** predicts a Net Power Score from six turbine parameters $(x_1, \dots, x_6)$, and **var2** predicts a Thermal Anomaly Score from three spatial coordinates (x_1, x_2, x_3) . Each training set has 1000 samples and each test set has 1000 samples. Only polynomial regression was used, as required.

2 Method

Both problems use the same pipeline:

1. Expand the inputs into all polynomial terms up to total degree d (the sum of the powers in each term is at most d), without a separate bias column.
2. Standardise every polynomial term to zero mean and unit variance. This keeps high-degree terms numerically stable and makes the regularisation penalty treat all terms comparably.
3. Fit a regularised linear model on the standardised terms.

The degree d was chosen by 5-fold shuffled cross-validation (CV) on the training data, with all preprocessing inside the pipeline so nothing leaks between folds. Every degree from 1 to 10 (var1) and 1 to 20 (var2) was tried. I picked the lowest degree whose mean CV MSE is within one standard error of the minimum, which prefers the simpler model when several degrees are statistically tied. The two regularisers were

$$\begin{aligned}\text{Ridge: } \min_w \|y - \Phi w\|_2^2 + \alpha \|w\|_2^2, \\ \text{Lasso: } \min_w \frac{1}{2n} \|y - \Phi w\|_2^2 + \alpha \|w\|_1,\end{aligned}$$

where Φ is the standardised polynomial feature matrix. Ridge shrinks all coefficients and suits problems where most terms carry some signal. Lasso sets many coefficients exactly to zero and suits problems where only a subset of terms is active.

3 Data Observations

In both datasets, roughly 25–30% of the values of each input equal exactly -1 or $+1$, which suggests the inputs were clipped to $[-1, 1]$. Clipping makes the observed relationship slightly different from a pure polynomial, which probably puts a floor under the achievable error. The target ranges are about $[-10.4, 12.4]$ for var1 and $[-29.3, 39.3]$ for var2.

4 Problem 1 (var1): Turbine Optimisation

With six inputs, a degree- d expansion has $\binom{d+6}{6} - 1$ terms (209 at $d = 4$, 461 at $d = 5$, 8007 at $d = 10$), which is large compared with 1000 samples, so regularisation is essential. With Lasso, CV MSE falls from 9.06 at degree 1 to 0.362 at degree 5 and then rises slowly (0.374 at degree 6, 0.421 at degree 10), the usual mild overfitting pattern (Figure 1). The gain from degree 4 (0.687) to degree 5 is much larger than the fold standard deviations, so degree 5 is clearly supported. Lasso keeps 112 of the 461 terms, so the underlying function appears sparse. At degree 5, $\alpha = 0.01$ was the best of five values tried ($\alpha = 0.001, 0.003, 0.01, 0.03, 0.1$ gave CV MSE 0.511, 0.389, 0.362, 0.408, 0.825). The final model is **degree 5 with Lasso** ($\alpha = 0.01$).

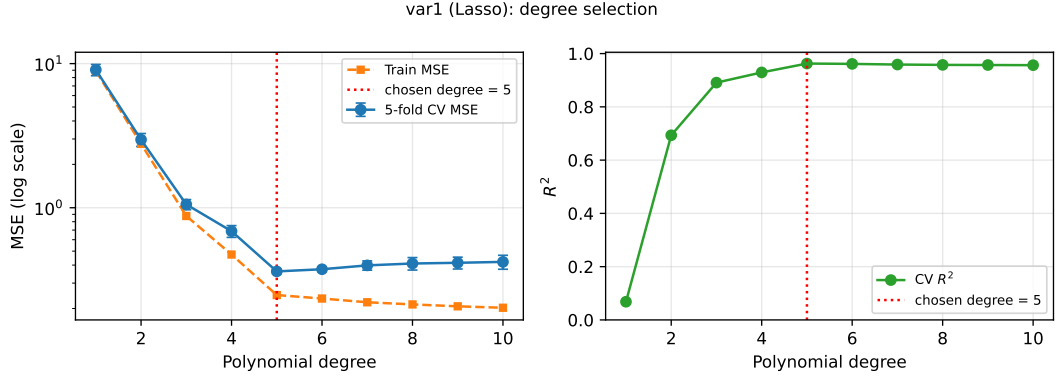


Figure 1: var1: CV MSE (error bars = ± 1 SD over folds), training MSE, and CV R^2 against degree.

5 Problem 2 (var2): Thermal Reservoir Mapping

With three inputs, degree 10 has 285 terms and degree 20 has 1770. CV MSE falls from 32.1 at degree 1 to 0.324 at degree 7 and then flattens (Figure 2). The minimum is at degree 12 (0.2431), but degrees 8–16 lie between 0.243 and 0.255, within about one fold standard deviation of each other, while training error keeps falling. The one-standard-error rule therefore selects the simpler **degree 10**, fitted with Ridge and $\alpha \approx 0.32$ chosen by RidgeCV. Beyond degree 16 the CV error rises (0.285 at degree 20), so very high degrees overfit even with Ridge. Lasso did not beat Ridge here.

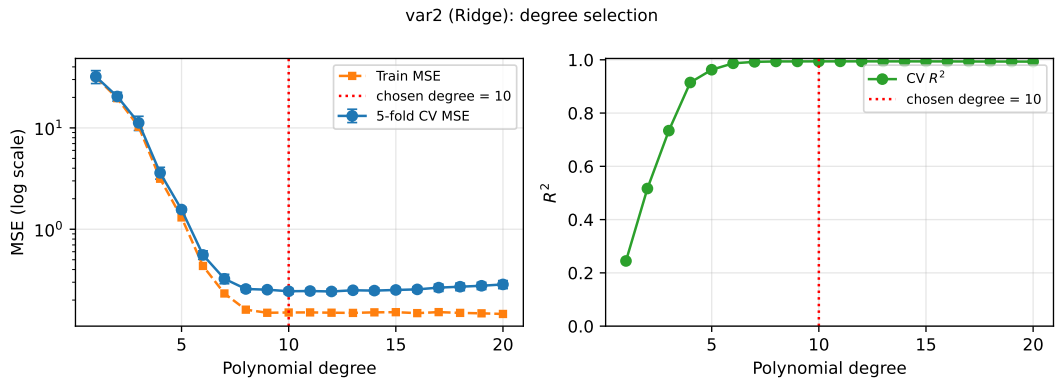


Figure 2: var2: CV MSE (error bars = ± 1 SD over folds), training MSE, and CV R^2 against degree.

6 Results

Table 1 lists the CV scores for every degree tried, and Table 2 summarises the final models. The test targets are hidden, so CV is the estimate of test performance. As a sanity check, each chosen

model was also trained on a random 80% of the training data and scored on the other 20%; the hold-out scores match the CV scores closely.

Table 1: 5-fold CV MSE (mean $\pm$ SD) and R^2 for every degree tried. Chosen degrees in bold.

Degree	var1 (Lasso)		var2 (Ridge)	
	CV MSE	CV R^2	CV MSE	CV R^2
1	9.0561 $\pm$ 0.8160	0.0683	32.0664 $\pm$ 4.6692	0.2447
2	2.9671 $\pm$ 0.3064	0.6934	20.4211 $\pm$ 2.0456	0.5166
3	1.0543 $\pm$ 0.0853	0.8910	11.2229 $\pm$ 1.7948	0.7340
4	0.6872 $\pm$ 0.0645	0.9293	3.6003 $\pm$ 0.4827	0.9148
5	0.3617 $\pm$ 0.0128	0.9626	1.5642 $\pm$ 0.0822	0.9627
6	0.3743 $\pm$ 0.0136	0.9613	0.5578 $\pm$ 0.0539	0.9868
7	0.3982 $\pm$ 0.0295	0.9589	0.3244 $\pm$ 0.0345	0.9923
8	0.4103 $\pm$ 0.0408	0.9576	0.2573 $\pm$ 0.0181	0.9938
9	0.4150 $\pm$ 0.0391	0.9571	0.2531 $\pm$ 0.0126	0.9940
10	0.4211 $\pm$ 0.0467	0.9565	0.2445 $\pm$ 0.0159	0.9942
11	–	–	0.2456 $\pm$ 0.0143	0.9941
12	–	–	0.2431 $\pm$ 0.0145	0.9942
13	–	–	0.2499 $\pm$ 0.0145	0.9941
14	–	–	0.2480 $\pm$ 0.0160	0.9941
15	–	–	0.2516 $\pm$ 0.0143	0.9940
16	–	–	0.2549 $\pm$ 0.0159	0.9939
17	–	–	0.2653 $\pm$ 0.0219	0.9937
18	–	–	0.2709 $\pm$ 0.0229	0.9935
19	–	–	0.2767 $\pm$ 0.0224	0.9934
20	–	–	0.2854 $\pm$ 0.0258	0.9932

Table 2: Final models.

Problem	Model	Degree	CV MSE	CV R^2	Hold-out MSE	Hold-out R^2
var1	Lasso	5	0.362	0.963	0.358	0.965
var2	Ridge	10	0.244	0.994	0.222	0.994

7 Other Techniques and Things Tried

- Plain least squares at all degrees (overfits at high degree).
- Ridge regression with a grid of α for every degree, and Lasso with a grid of α (including a variant that refits the Lasso-selected terms by least squares, which was not better).
- Repeating cross-validation with a different random shuffle to confirm that the scores are stable (differences of about 0.01 in MSE).

8 Rationale and Discussion

- **Bias–variance trade-off.** Below the true complexity the polynomial cannot represent the surface (high bias); above it, extra monomials fit noise (high variance). The CV curves show this U-shape, and the selected degrees sit at the start of the flat or rising part.
- **Why cross-validation.** Training error always falls with degree and cannot reveal overfitting (see the training-MSE curves). CV estimates performance on unseen points and so mirrors the hidden test evaluation.

- **Why regularisation.** With hundreds of thousands of correlated terms and only 1000 samples, unregularised least squares is ill-conditioned. Regularisation keeps the fit stable at higher degrees and makes the learned structure meaningful.
- **Why the chosen degrees are below the stated maximum.** The problem statements give upper bounds on the true degree (10 and 20). The data support lower effective degrees: higher-order terms are small relative to the noise and clipping, and including them does not lower CV error.
- **Why different regularisers.** Lasso gave the lowest error for the six-variable problem, which suggests many monomials are inactive. Ridge suits the dense, smooth 3D field in var2.

Limitations. Polynomials can extrapolate poorly near the edges of the input box. For var1, 8 of the 1000 test predictions exceed the largest training target (maximum predicted 17.1 versus 12.4 in training); these come from corner regions where training data are sparse. I did not clip them because the true values could lie outside the training range. The error floor of about 0.24 (var2) and 0.36 (var1) also likely reflects the clipping of inputs and noise in the targets rather than a choice of degree.

9 Conclusion

For var1 a sparse degree-5 polynomial fitted with Lasso gives CV $R^2 \approx 0.963$. For var2 a degree-10 polynomial fitted with Ridge gives CV $R^2 \approx 0.994$. Predictions were written to `BT2024136_pred_var1.csv` and `BT2024136_pred_var2.csv`.